

# VEHICLE PARKING APP

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I am a passionate computer science student interested in full-stack development and problem solving. I enjoy building real-world applications and learning through practical projects like this one.

## **DESCRIPTION:**

This project is a Vehicle Parking App built using Flask. It allows users to reserve parking spots and enables admins to manage lots, spots, and view analytics. The project focuses on authentication, role-based access, and basic CRUD operations.

## **TECHNOLOGIES USED:**

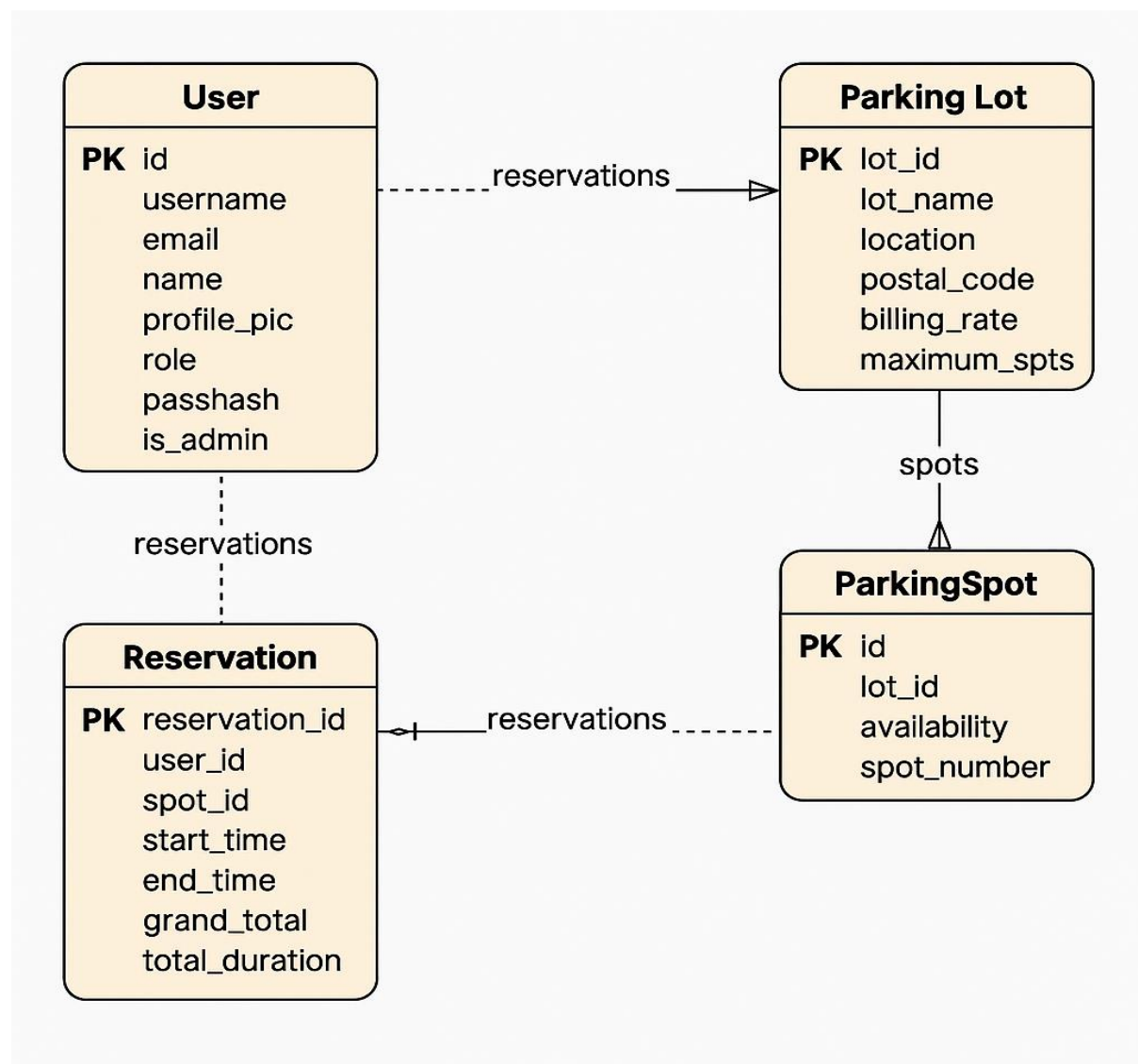
- FLASK : Web framework to handle routing and app logic
- FLASK\_SQLAlchemy : ORM to interact with the SQLite database
- WERKZEUG.SECURITY : User authentication
- JINJA2 : Template engine for rendering HTML pages
- HTML/BOOTSTRAP : Frontend design and layout
- CHART.JS : Implementing summaries with pictorial representation
- DOTENV : To manage secret keys securely
- GIT : For development and version control

## **AI PERCENTAGE:**

|                       |     |
|-----------------------|-----|
| HTML + JINJA2         | 15% |
| Bootstrap             | 8%  |
| Charts                | 3%  |
| Testing and Debugging | 1%  |

## DB SCHEMA DESIGN:

The schema models core entities—User, Admin, ParkingLot, ParkingSpot, and Reservation—with clearly defined foreign key relationships to ensure referential integrity. It follows normalization principles (up to 3NF), minimizing data redundancy and maintaining consistency. One-to-many and many-to-one mappings are implemented using SQLAlchemy ORM relationships for efficient data traversal. The design enables robust role-based access control and supports key operations like spot allocation, availability tracking, and historical reservation queries. The structure is optimized for performance and future extensibility.



## VIDEO:

[https://drive.google.com/file/d/1ovuLJw\\_eCxTk0aMa-L4P4Y1aXbDUPGFb/view?usp=sharing](https://drive.google.com/file/d/1ovuLJw_eCxTk0aMa-L4P4Y1aXbDUPGFb/view?usp=sharing)